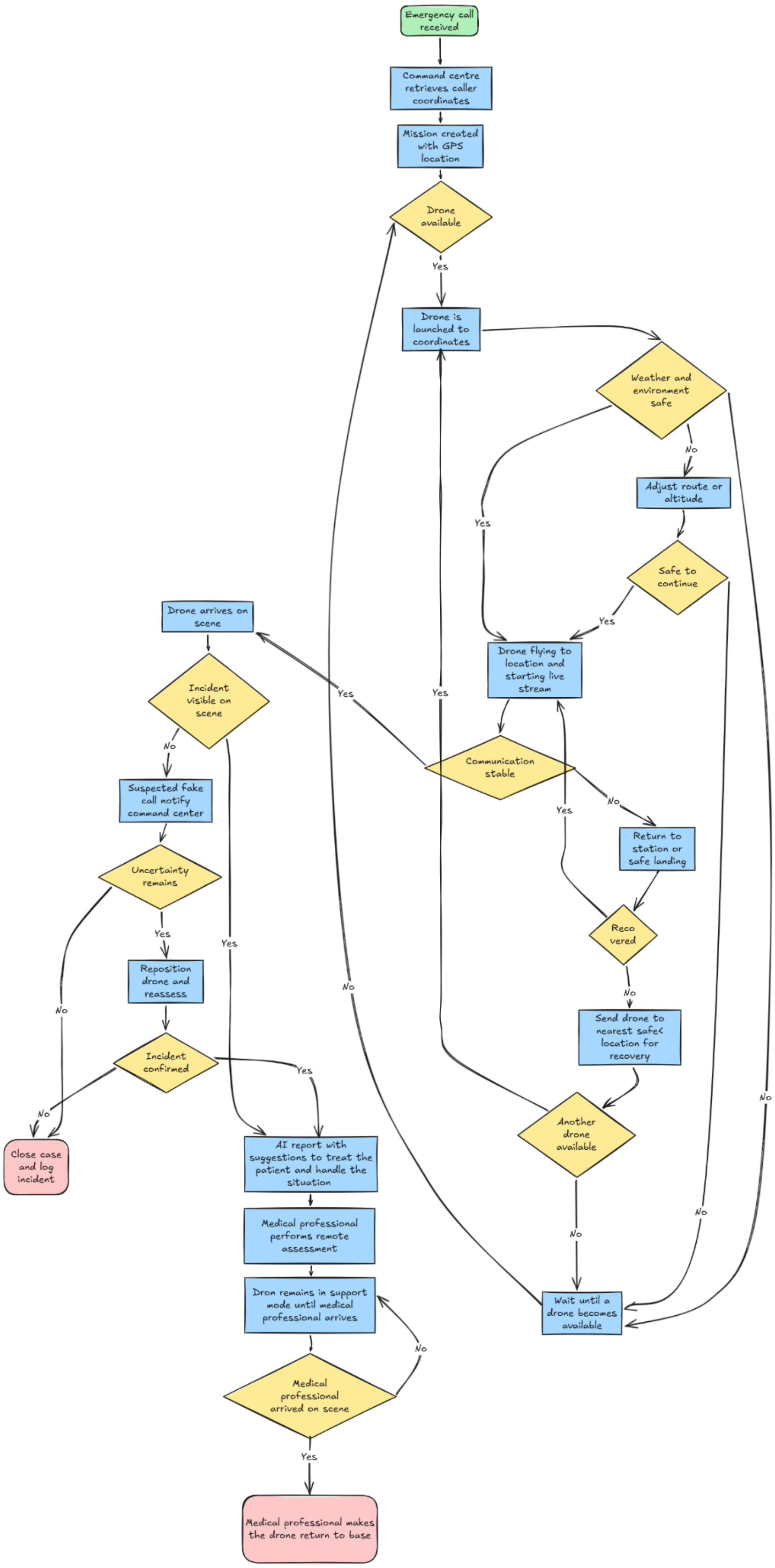
# Crisis management protocol

- Flowchart to determine the deployment of drones



# Protocol Structure – Disaster Response & Crisis Management

### 4.1 What to Do?

1. Receive alert of a medical incident
2. Deploy drone to incident location
3. Perform aerial assessment and live video streaming
4. AI report of the incident
5. Support on-site response until EMS arrival

### 4.2 When to Do What?

**Phase 1 – Incident receival**

- Triggered by emergency call, security report.

**Phase 2 – Drone Deployment (0–2 minutes)**

- Immediate launch after receiving phone call

**Phase 3 – Remote Assessment (2–5 minutes)**

- Visual confirmation of:
  - Injury and number of the victims
  - Consciousness and mobility
  - Environmental risks

**Phase 4 – EMS Coordination (parallel)**

- Information relayed to paramedics and hospitals
- Route optimization provided

**Phase 5 – Handover**

- Drone remains in support mode until EMS confirms control

### 4.3 Who Is Responsible?

- **Command Center**

    - Medical specialists
    - Central communication hub
- **Medical Coordinator (EMS Lead)**
    - Interprets drone visuals medically
    - Determines response priority
    - Responsible for drone instructions
- **Paramedic Teams**
    - On-ground medical intervention
- **Hospitals**
    - Prepare for incoming patients based on early assessment

### 4.4 How to Communicate?

- **Visual Communication**
    - Live HD video feed to command center and medical coordinator
- **Public Communication**
    - Drone loudspeaker used when giving instructions
    - Clear, short, and calm instructions to bystanders
- **Data Handling**
    - Video data used strictly for medical and safety purposes
    - Compliance with privacy and data protection regulations

### 4.5 Which Procedures to Follow?

- Standardized medical assessment checklist for specialist in command centre.
- Escalation matrix for minor vs. critical incidents
- Emergency landing and fail-safe procedures
- AI learning after incident??

## Communication Flow Overview

**Alert → Drone → Command Center(professional) → Bystander → Hospital (if required)**
Parallel: Controlled communication to bystanders/public when necessary.

# Communication Plan

### 1) Receive Alert of a Medical Incident

**Trigger Sources**
The alert can be triggered by an emergency call (112) or by a report from security personnel.

**Who Communicates?**
The reporting party communicates directly with the professional in the Command Center. At the same time, the AI system informs the intervening paramedics.

**How?**
Communication takes place via a secure radio network and through digital report notifications.

**Information Shared:**
The exact location (GPS), the type of incident (if known), and the estimated number of victims are communicated immediately. In addition, information about immediate risks such as crowd density or other hazards is shared, together with an initial indication of the severity of the injuries.

**Responsibility:**
The professional in the Command Center validates the incident. This professional also provides instructions to bystanders when necessary. Meanwhile, the AI generates an analysed report of the incident to support the response process.

### STEP 2: Deploy Drone to Incident Location

**Who Communicates?**
The emergency call system automatically deploys the drone.

**How?**
This is done via a direct communication line.

**Information Shared:**
The system analyses the fastest route for responders to reach the incident location.

**STEP 3: Perform Aerial Assessment & Live Video Streaming and Support Bystanders (On-Site Response)**

**Who Communicates?**
The drone transmits information to the Command Center. The Command Center communicates with the medical professional, and the live video feed is shared with this professional. The medical professional then communicates directly with bystanders on site. They make suggestions about how to handle the situation and how to treat the patient if needed.

**How?**
A secure live HD video stream is used, along with a dedicated medical radio channel.

**Information Shared:**
Information includes the number of victims, the severity of the injuries, environmental hazards, the level of crowd accessibility, and specific instructions given to bystanders.

**STEP 4: AI Report of the Incident**

**Function:**
The AI system analyzes crowd density, victim movement, possible trauma indicators, and route optimization for emergency responders.

**Who Receives AI Report?**
The Command Center and the emergency responders receive the AI report.

**How?**
The report is delivered through an integrated incident dashboard and via automated digital alerts.

**Communication Protocol:**
The AI output is considered a suggestion rather than a confirmed medical fact. The AI supports medical professionals but does not replace medical decision-making.

**Information Forwarded to Responding Forces:**
Responders receive the severity category (Minor / Serious / Critical → hospital), the suggested access route, and any relevant risk warnings.